

CAI1001C

Data Visualization & Intro to Machine Learning

Weeks 5 & 6 Combined Session

► Miami Dade College ► Prof. [Name]

Tonight's Agenda

ROADMAP FOR THE SESSION

01

Data Visualization

OVERVIEW & BEST PRACTICES

02

Intro to Machine Learning

DEEP DIVE: CONCEPTS & WORKFLOW

03

Live Demo

CODING IN GOOGLE COLAB

04

Group Lab

HANDS-ON PRACTICE & ASSIGNMENT

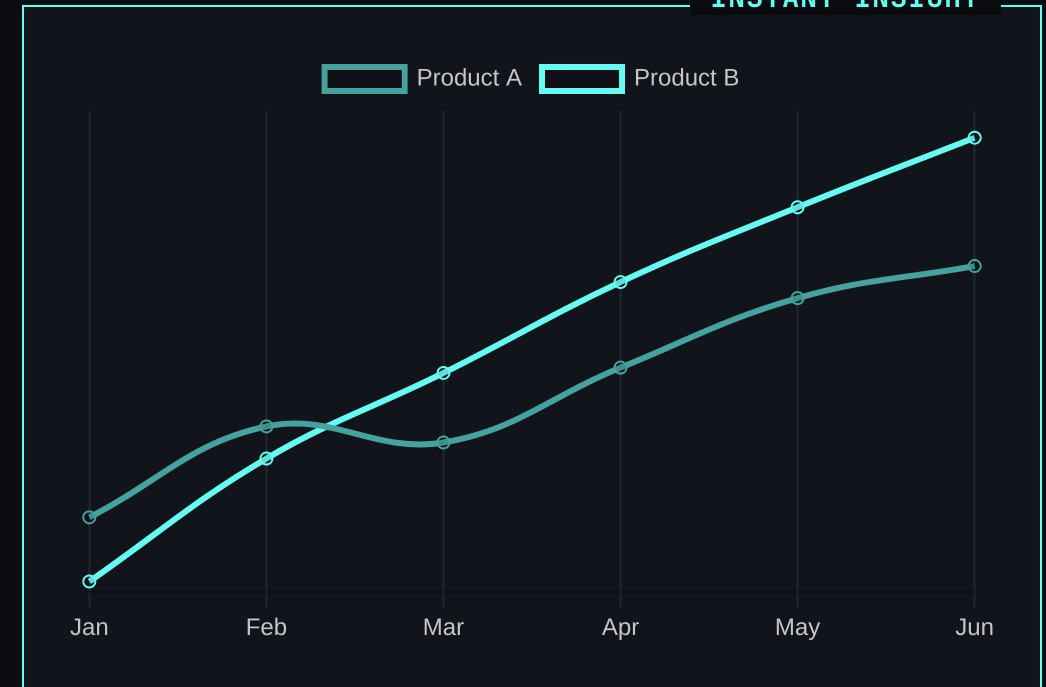
Why Visualize Data?

FROM RAW NUMBERS TO INSTANT INSIGHT

RAW DATA (HARD TO READ)

Month	Rev (A)	Rev (B)
Jan	12,400	11,200
Feb	14,100	13,500
Mar	13,800	15,100
Apr	15,200	16,800
May	16,500	18,200
Jun	17,100	19,500

INSTANT INSIGHT



“A good chart communicates in **seconds** what a spreadsheet takes **minutes** to explain.”

The Golden Rule

QUESTION FIRST, CHART SECOND

The question determines the chart type,
not the other way around.

"How do sales compare across stores?"



BAR CHART

"How has revenue changed over time?"



LINE CHART

"What's the grade distribution?"



HISTOGRAM

"Are study hours and GPA related?"



SCATTER PLOT

Chart Types Quick Reference

CHOOSING THE RIGHT TOOL



Bar Chart

Comparing categories



Line Chart

Trends over time



Histogram

Distribution of one variable



Scatter Plot

Relationship between two variables



Pie Chart

Parts of a whole

USE SPARINGLY!



Heatmap

Patterns across many variables

Two Python Libraries

THE FOUNDATION VS. THE UPGRADE

Matplotlib

The foundation. Full control over every pixel, but requires more code.

```
import matplotlib.pyplot as plt

plt.plot(x, y)
plt.title("My Chart")
plt.show()
```

Seaborn

Built on Matplotlib. Prettier defaults, statistical plots, less code.

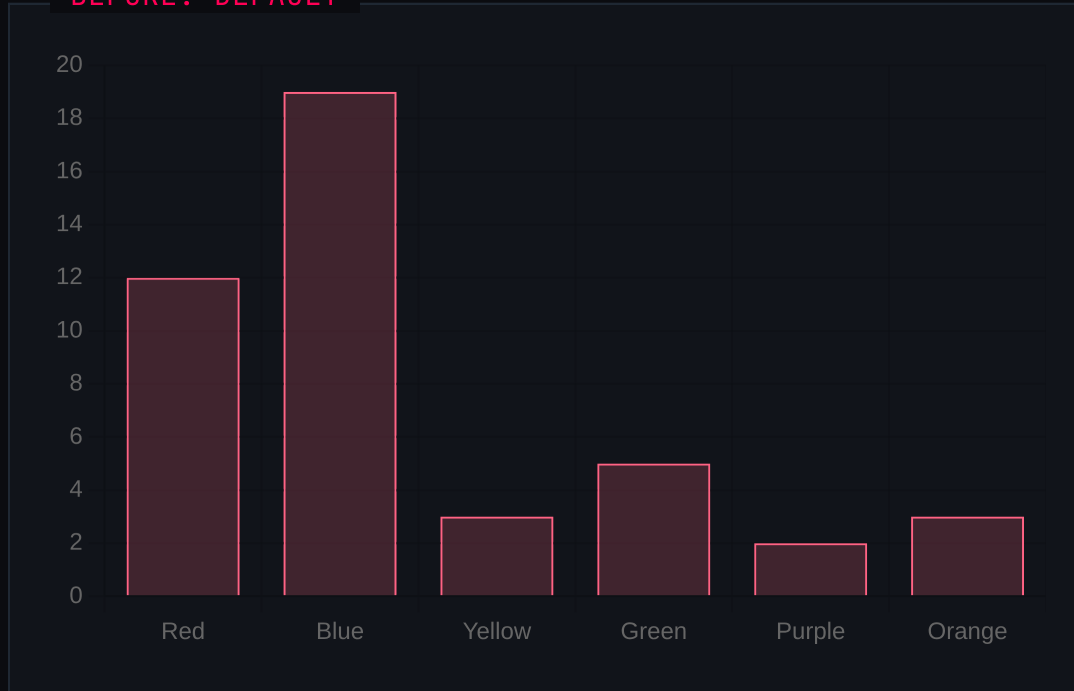
```
import seaborn as sns

sns.lineplot(data=df, x="year", y="sales")
# Automatic themes & labels!
```

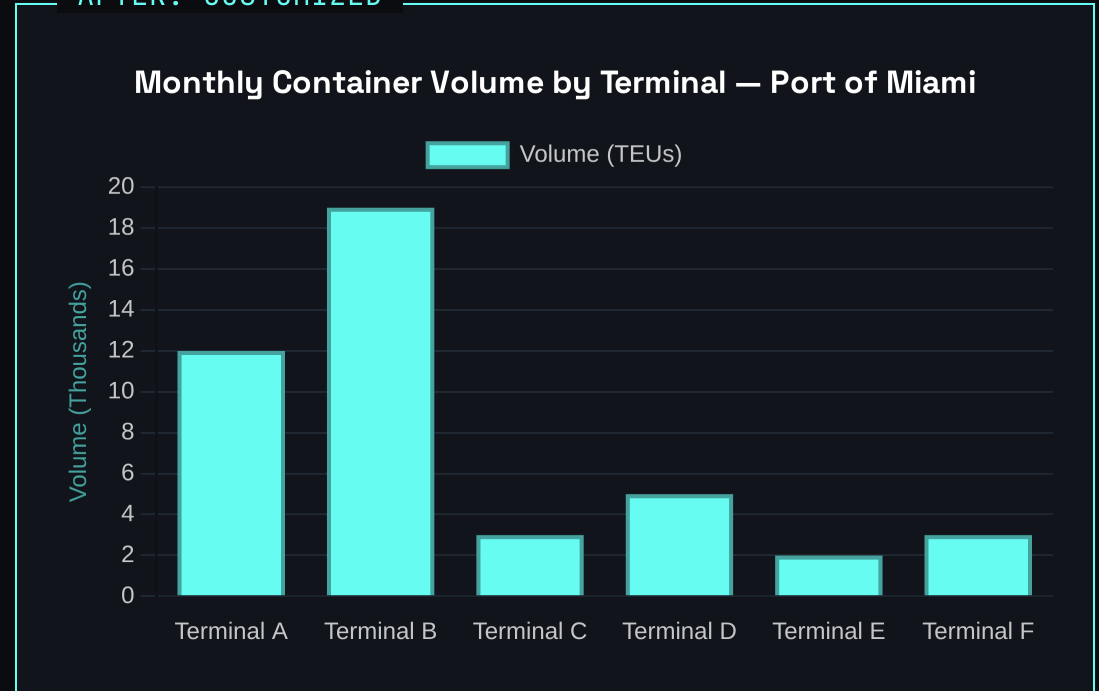
From Raw to Readable

CUSTOMIZATION MATTERS

BEFORE: DEFAULT



AFTER: CUSTOMIZED



"A chart without labels is like a map without street names."

Watch Out: Misleading Charts

DON'T LIE WITH DATA



Truncated Y-Axis

Starting the axis at 95 instead of 0 makes a 1% difference look like a 50% crash.



Cherry-Picked Ranges

Showing only the "good months" while hiding the overall downward trend.



3D Distortion

3D pie charts make the front slice look huge and the back slice look tiny, regardless of data.



Dual Y-Axes

Plotting "Ice Cream Sales" and "Shark Attacks" on the same chart to imply a false correlation.

Key Takeaway: Charts can lie. Your job is to make **honest** ones.

Data Viz Recap

PART A COMPLETE

- ✓ Start with a question, not a chart type.
- ✓ Pick the right tool (Bar, Line, Scatter, Hist).
- ✓ Label everything clearly (Titles, Axes, Legends).
- ✓ Be honest—don't distort the data.

Next: We've learned to **see** patterns.
Now, let's teach computers to **learn** them.

PART B

Introduction to Machine Learning

From Seeing Data to Learning From It

What Is Machine Learning?

A NEW WAY OF PROGRAMMING

TRADITIONAL PROGRAMMING

Rules

+

Data



Answers

MACHINE LEARNING

Answers

+

Data



Rules

Instead of writing the rules, we teach the computer to **find** them.

The Three Types of ML

HOW MACHINES LEARN



Supervised

"THE TEACHER"

Learning with labeled data. We give the answer key.

 Spam Filter



Unsupervised

"THE EXPLORER"

Finding patterns in unlabeled data. No answer key.

 Customer Segments



Reinforcement

"THE GAMER"

Learning through trial and error. Action & Reward.

 Chess Bot

Supervised Learning

THE TWO MAIN TASKS



Regression

"How much?"

Predicting a continuous numerical value.

 House Prices

 Temperature

 Stock Value



Classification

"Which one?"

Predicting a category or label.

 Spam vs. Not Spam

 Cat vs. Dog

 Disease: Yes/No

The ML Workflow

FROM RAW DATA TO PREDICTION



1. Collect

Gather raw data



2. Prep

Clean & define features



3. Split

Train vs. Test sets



4. Train

Model learns patterns



5. Evaluate

Test accuracy



6. Predict

Use on new data

Features and Labels

THE VOCABULARY OF ML

CREDIT SCORE	INCOME	DEBT LEVEL	LOAN APPROVED?
720	\$85k	Low	Yes
640	\$45k	High	No
800	\$120k	Medium	Yes
Features (X)			Label (y)

Features (X)

The input data used to make a prediction.

Label (y)

The output or answer we are trying to predict.

Train/Test Split

WHY IT MATTERS

Training Set (80%)

Test Set (20%)



The Textbook

Data the model studies to learn patterns.



The Exam

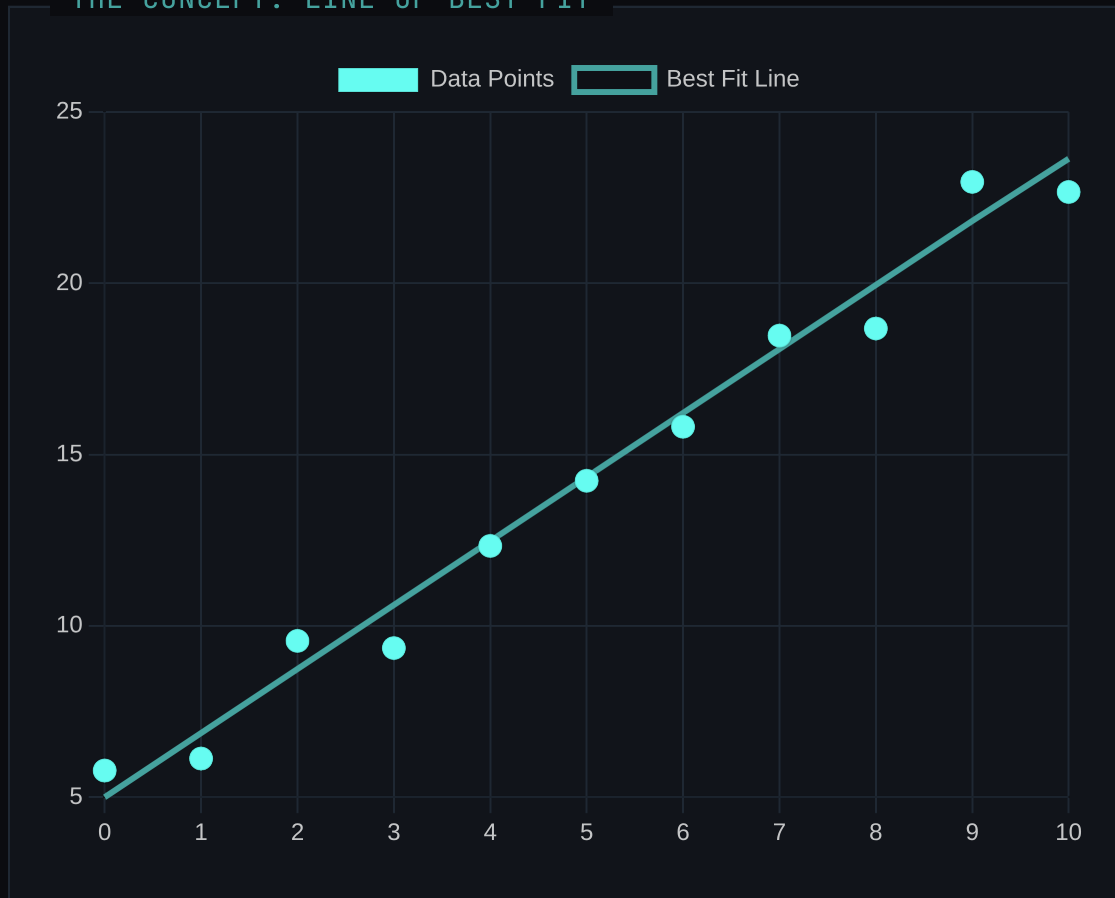
New data to check if it learned.

We hide the test data to prevent **memorization** (overfitting).

Linear Regression

YOUR FIRST ALGORITHM

THE CONCEPT: LINE OF BEST FIT



1. Import & Create

```
from sklearn.linear_model import  
LinearRegression  
model = LinearRegression()
```

2. Train (Fit)

```
# Learn the pattern from training data  
model.fit(X_train, y_train)
```

3. Predict

```
# Make predictions on new data  
predictions = model.predict(X_test)
```

Reading the Results

TWO KEY METRICS



Coefficients

THE INFLUENCE

Tells you how much each feature affects the prediction. Positive means it increases the result; negative means it decreases it.

$$y = mx + b$$

Example: +\$5,000 salary per year of experience



R-Squared

THE ACCURACY

A score from 0 to 1. Tells you how well the model fits the data.

1.0 = Perfect Fit

0.0 = Random Guessing

0.85



When "Accurate" Isn't Fair

THE HIDDEN BIAS PROBLEM

The Accuracy Trap

A model can be 95% accurate overall but 60% accurate for certain groups.

Skewed Data

If training data is 80% male, the model learns that "male" is a feature of success.

REAL WORLD CASE

Amazon's Hiring AI

Trained on 10 years of resumes (mostly men). It learned to penalize resumes containing the word "women's" (e.g., "Women's Chess Club").

"The model wasn't broken. It did exactly what the data taught it."

The Questions You Should Always Ask

CRITICAL THINKING FOR AI



Whose data trained this model?



Who benefits from the prediction? Who gets hurt?



Is "accurate on average" good enough?

REAL WORLD EXAMPLE

Loan approval models in Miami: If training data underrepresents small family-owned businesses, what happens to **Sofia's restaurant?**

This Connects to Your Friday Assignment

DON'T JUST DESCRIBE THE TOOL

“What data built this?
Who does it work well for?
Who might it **miss**?”

There's no right answer — I want your honest take.

NEXT CLASS



Neural Networks & Deep Learning

Mimicking the Human Brain

Mission Debrief

KEY TAKEAWAYS



Data Viz: See the pattern first.



ML: Teach the computer to learn it.



Supervised: Learning with an answer key.



Split: Don't memorize (overfit).



Ethics: Bias In = Bias Out. Question everything.

Questions?